REMARKS ON THE ZEROS OF OPTIMAL POLYNOMIAL APPROXIMANTS

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ABSTRACT. Let q_n be the optimal polynomial approximants of $1/f$ in a weighted Hardy space H^2_ω. Bénéteau and Centner asked whether, for cyclic f and ζ on the unit circle, some sector with vertex at ζ is free of zeros of the q_n. We show that in H^2 the zeros for $f(z) = 1 - z$ enter every sector at $\zeta \neq 1$ which points out of the disc, and we construct an admissible weight ω for which no sector at any $\zeta \neq 1$ is zero-free when $f(z) = (1 - z)^2$. Orive, Sánchez-Lara and Seco conjectured that the spurious zeros of the orthogonal polynomials φ_n on the unit circle associated with a critical polynomial can be paired with the zeros of their electrostatic partners S_n, with an explicit displacement of order $1/n$. We show that this fails for $f(z) = (1 - z)(z^2 - 2z\cos\theta + 1)$ and a suitable θ: along a subsequence two spurious zeros collide on a scale finer than $1/n$, and the two zeros of S_n are displaced by 0 and $2\xi/n$ instead of ξ/n. Finally, we note that the asymptotics of Martínez-Finkelshtein, McLaughlin and Saff imply the conjectured weak limit of the zeros of the optimal polynomial approximants for positive powers of critical polynomials.

1. INTRODUCTION

Let $\omega = (\omega_k)_{k\geq 0}$ be a sequence of weights with

$$\omega_0 = 1, \qquad \omega_k > 0, \qquad \lim_{k\to\infty} \frac{\omega_k}{\omega_{k+1}} = 1, \tag{1}$$

and let H^2_ω denote the space of analytic functions $f(z) = \sum_{k\geq 0} a_k z^k$ in the unit disc $\mathbb{D}$ for which

$$\|f\|^2_\omega = \sum_{k=0}^{\infty} |a_k|^2 \omega_k < \infty.$$

The choice $\omega_k = (k + 1)^\alpha$ gives the Dirichlet-type spaces D_α, among them the Hardy space $H^2 = D_0$. Let $\mathcal{P}_n$ be the polynomials of degree at most n. For $f \in H^2_\omega$, $f \not\equiv 0$, the nth *optimal polynomial approximant* (opa) of $1/f$ is the polynomial $q_n \in \mathcal{P}_n$ which minimizes $\|pf - 1\|_\omega$ over $p \in \mathcal{P}_n$. In other words, $q_n f$ is the orthogonal projection of 1 onto $f\mathcal{P}_n$. A function

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f is *cyclic* if its polynomial multiples are dense in H^2_ω, which happens if and only if $\|q_n f - 1\|_\omega \to 0$. We follow the notation of the survey of Bénéteau and Centner [2], whose open questions we cite with the numbering of arXiv:2102.01725v1.

The zeros of opas have been studied by Bénéteau, Khavinson, Liaw, Seco and Sola [4] and by Bénéteau, Khavinson, Liaw, Seco and Simanek [5]. In the second paper the smallest possible modulus of a zero is identified with the norm of a Jacobi matrix J_ω; in particular, opas never vanish in $\mathbb{D}$ when $\|J_\omega\| = 2$, and they never vanish in $\overline{\mathbb{D}}$ when ω is nondecreasing [5, Corollary 4.3]. For weights whose norms are equivalent to those of regular rotation invariant measures (*quasi-integral* weights) and cyclic f such that $1/f$ has a singularity on $\mathbb{T}$, along a subsequence almost all zeros of q_n lie in the disc of radius $1+\varepsilon$ for every $\varepsilon > 0$, and every point of $\mathbb{T}$ is a limit point of zeros [5, Theorems 6.1 and 6.2], in analogy with Jentzsch's theorem for Taylor polynomials. Bénéteau and Centner then asked for more precise information near the circle.

Question ([2, Open Question 2])**.** *Given a cyclic function $f \in H^2_\omega$, and given $\zeta \in \mathbb{T}$, is there a sector with vertex at ζ that is devoid of zeros of opas? More generally, are there regions or rays that the zeros avoid?*

To state our results, we write

$$S(\zeta; \theta_0, \delta, \rho) = \{\zeta(1 + re^{i\theta}) : \ 0 < r < \rho, \ |\theta - \theta_0| < \delta\}$$

for $\zeta \in \mathbb{T}$, $\theta_0 \in \mathbb{R}$, $0 < \delta < \pi$ and $\rho > 0$. Directions are measured from the outward normal ζ, so the direction θ points out of $\overline{\mathbb{D}}$ when $\cos\theta > 0$ and into $\mathbb{D}$ when $\cos\theta < 0$. Every sector with vertex at ζ, bounded or an unbounded cone, contains some $S(\zeta; \theta_0, \delta, \rho)$ with $\cos\theta_0 \neq 0$.

Our first result describes the simplest example completely. Part (a) is due to Bénéteau, Khavinson, Liaw, Seco and Simanek, and we include it for comparison.

Theorem 1. *Let $\omega_k = 1$ for every k, so that $H^2_\omega = H^2$, let $f(z) = 1 - z$, and let $\zeta \in \mathbb{T} \setminus \{1\}$.*

(a) *Every zero of every q_n lies in $\mathbb{C}\setminus\overline{\mathbb{D}}$. Hence, if $\cos\theta < 0$ for $|\theta - \theta_0| \leq \delta$, then $S(\zeta; \theta_0, \delta, \rho)$ contains no zero of any q_n when ρ is small enough.*

(b) *Suppose that $\cos\theta_0 > 0$. Then for every $\delta > 0$ and $\rho > 0$, the sector $S(\zeta; \theta_0, \delta, \rho)$ contains a zero of q_n for every sufficiently large n.*

The argument for (a) works for every cyclic f in every space with a nondecreasing weight, such as D_α with $\alpha \geq 0$: small sectors which point into the disc are always zero-free. So in these spaces the question is only interesting for the remaining directions, and (b) shows that for $f(z) = 1 - z$ in H^2 none of them is zero-free. In particular, if sectors are understood as unbounded cones, then no cone with vertex at $\zeta \neq 1$ is zero-free in this example. Indeed, a cone pointing into the disc leaves $\overline{\mathbb{D}}$ through an open arc of $\mathbb{T}$, and it is a neighbourhood of each point $\zeta' \neq 1$ of that arc, so it contains a small sector at ζ' pointing out of the disc, which contains zeros by (b).

The main purpose of the first half of the present paper is to show that in the generality of the survey [2], where ω is any weight satisfying (1), the answer is no, even for sectors pointing into the disc. Let $M_0 = 0$ and $M_m = 4^m$ for $m \geq 1$. Put $\varepsilon_m = M_m^{-1/4}$ for even m and $\varepsilon_m = M_m^{-3/4}$ for odd m, and define $\omega_0 = 1$ and

$$\omega_k = \omega_{M_{m-1}} e^{-\varepsilon_m (k - M_{m-1})}, \qquad M_{m-1} < k \leq M_m, \quad m \geq 1. \tag{2}$$

Then ω is nonincreasing, and $\omega_k/\omega_{k+1} = e^{\varepsilon_m}$ for $M_{m-1} \leq k < M_m$. Since $\varepsilon_m \to 0$, the weight (2) satisfies (1).

Theorem 2. *Let ω be the weight (2) and $f(z) = (1-z)^2$. Then f is cyclic in H^2_ω. Suppose that $\zeta \in \mathbb{T} \setminus \{1\}$, that $\cos\theta_0 \neq 0$, and that $\delta, \rho > 0$. Then $S(\zeta; \theta_0, \delta, \rho)$ contains a zero of $q_{M_m - 2}$ for every sufficiently large m which is odd if $\cos\theta_0 > 0$ and even if $\cos\theta_0 < 0$.*

Consequently, for this ω and f, no sector with vertex at a point $\zeta \neq 1$ is devoid of zeros of opas, and every open set which meets $\mathbb{T} \setminus \{1\}$ contains zeros of infinitely many q_n. We comment on the regions and rays in the second half of the question in Section 6.

The second half of the paper concerns the Hardy space and a conjecture of Orive, Sánchez-Lara and Seco [8], whose results we cite with the numbering of arXiv:2507.15488v1. In H^2 we follow their notation and write p_n for the opa of $1/f$. Let φ_n be the monic orthogonal polynomial of degree n for the measure $|f|^2 \, \mathrm{d}m$ on $\mathbb{T}$, where m is normalized arc length. Then p_n is a constant multiple of the reversed polynomial $\varphi_n^*(z) = z^n \overline{\varphi_n(1/\overline{z})}$ [8, (2)], so the zeros of p_n are the reflections $1/\overline{z}$ of the zeros z of φ_n. A *critical polynomial* is a polynomial without zeros in $\mathbb{D}$ and with at least one zero on $\mathbb{T}$. For such f, most zeros of φ_n approach $\mathbb{T}$, but a bounded number of them, the *spurious zeros*, may stay in a compact subset of $\mathbb{D}$; by a theorem of Martínez-Finkelshtein, McLaughlin and Saff [7, Corollary 1] there are at most $m-1$ of them when f has m distinct zeros on $\mathbb{T}$. Orive, Sánchez-Lara and Seco interpret the zeros of φ_n as an electrostatic equilibrium, in which the spurious zeros are held in place by attractive charges at the zeros of the *electrostatic partner* S_n of φ_n, a polynomial of degree at most $m-1$ which we recall in Section 4. Based on numerical evidence and a heuristic argument, they made the following conjecture.

Conjecture ([8, Conjecture 1])**.** *Let $\{\xi_{n,k}\}$ be the set of spurious roots of φ_n and $\{s_{n,\ell}\}$ the set of the roots of the electrostatic partner S_n which are inside the disk and do not accumulate on $\mathbb{T}$. Then both sets has the same amount of elements and we can reorganize all of them in pairs $(\xi_{n,k}, s_{n,k})$ in such a way that*

$$\xi_{n,k} = s_{n,k} - \frac{s_{n,k}}{n} + o(1/n),$$

for any of these pairs.

We show that the pairing fails for the critical polynomials

$$f_\theta(z) = (1-z)(z^2 - 2z\cos\theta + 1) = (1-z)(z-e^{i\theta})(z-e^{-i\theta}), \tag{3}$$

which have the three simple zeros 1 and $e^{\pm i\theta}$ on $\mathbb{T}$ and $f_\theta(0) = 1$. This is the example $(z^2 - 2z\cos\theta + 1)$ of [8, Section 4.1] with one more zero added at 1, which is the smallest change that allows two spurious zeros.

Theorem 3. *There is $\theta \in (0,\pi)$ with the following property. Let φ_n be the monic orthogonal polynomials for $|f_\theta|^2\,\mathrm{d}m$, and let S_n be their electrostatic partners. There are integers $n_j \to \infty$, real numbers $\eta_j \to \xi$ with $0 < \xi < 1/4$, and for each j two zeros $\xi_{j,1}, \xi_{j,2}$ of φ_{n_j} and two zeros $s_{j,1}, s_{j,2}$ of S_{n_j}, such that the following hold.*

(a) *For every $1/4 < r < 1$ and every sufficiently large j, $\xi_{j,1}$ and $\xi_{j,2}$ are the only zeros of φ_{n_j} in $\{|z| \le r\}$, and S_{n_j} has degree two, with zeros $s_{j,1}$ and $s_{j,2}$ in $\{|z| < r\}$.*

(b) *As $j \to \infty$,*

$$n_j(\xi_{j,1} - \eta_j) \to 0, \quad n_j(\xi_{j,2} - \eta_j) \to 0,$$
$$n_j(s_{j,1} - \eta_j) \to 0, \quad n_j(s_{j,2} - \eta_j) \to 2\xi.$$

In particular, for either bijection σ of $\{1,2\}$,

$$\lim_{j\to\infty} \max_{k=1,2} n_j \Big| \xi_{j,k} - s_{j,\sigma(k)} + \frac{s_{j,\sigma(k)}}{n_j} \Big| = \xi > 0. \tag{4}$$

By (a), along n_j the spurious zeros of φ_{n_j} are exactly $\xi_{j,1}, \xi_{j,2}$, and the zeros of S_{n_j} in $\mathbb{D}$ which do not accumulate on $\mathbb{T}$ are exactly $s_{j,1}, s_{j,2}$. Both sets have two elements, so the cardinality part of the conjecture holds in this example; it is the displacement which fails. Note that the two partner zeros are displaced from η_j by 0 and $2\xi/n_j$, and their average is displaced by ξ/n_j, as the conjecture predicts for each of them. The weight is fixed: θ does not depend on n.

Orive, Sánchez-Lara and Seco also discuss the weak asymptotics of the zeros of p_n. Acuaviva and Seco [1] proved that the zeros of $1 - p_n f$ are asymptotically uniformly distributed on $\mathbb{T}$, and Orive, Sánchez-Lara and Seco write: "It is conjectured that the same should hold for the OPA $\{p_n\}$ themselves" [8, p. 3]. Here f is a critical polynomial or a positive power of one. The final result of the present paper records that this follows from the work of Martínez-Finkelshtein, McLaughlin and Saff [7].

Theorem 4. *Suppose that $a_1, \dots, a_m$ are distinct points of $\mathbb{T}$, that $\beta_1, \dots, \beta_m$ are positive, and that g is analytic and without zeros in a neighbourhood of $\overline{\mathbb{D}}$. Let*

$$F(z) = g(z) \prod_{k=1}^{m} (1 - \overline{a_k} z)^{\beta_k},$$

with the branches equal to 1 at $z = 0$, and let p_n be the opa of $1/F$ in H^2. Then $\deg p_n \ge n - m + 1$ for all sufficiently large n, and the normalized zero counting measures of p_n converge weakly to normalized arc length on $\mathbb{T}$.

Every positive power $F = P^\gamma$ of a critical polynomial P has this form: P is a constant times $\prod_k (1 - \overline{a_k} z)^{m_k}$ times a polynomial without zeros on $\overline{\mathbb{D}}$, and we may take $\beta_k = \gamma m_k$. The argument is a routine combination of [7, Corollary 1 and Theorem 1(iii)] with the reversal $p_n \propto \varphi_n^*$; the only point to watch is that a few zeros of p_n may escape to infinity.

Let us briefly describe our approach. Both Theorem 1 (b) and Theorem 2 rest on a description of the residual $R = 1 - q_n f$, whose coefficients are u_k/ω_k with (u_k) constant for $f = 1 - z$ and affine in k for $f = (1-z)^2$. Near ζ we write $z = \zeta e^{s/M}$, where $M = n + 2$. Then $z^M = \zeta^M e^s$, and the equation $R(z) = 1$ for the zeros of q_n becomes $B_M e^s \approx 1$ for a number B_M which we compute. Its solutions form a comb: $\operatorname{Re} s \approx \lambda_M = -\log|B_M|$, with imaginary parts spaced by 2π. Hence the zeros near ζ lie close to the circle $\{|z| = e^{\lambda_M/M}\}$, at mutual distance about $2\pi/M$. A direction which is not tangent to $\mathbb{T}$ meets this circle at distance about $|\lambda_M|/M$ from ζ, where a sector of fixed opening has width about $|\lambda_M|/M \gg 2\pi/M$; so the sector contains a zero, provided that $|\lambda_M| \to \infty$ and that the sign of λ_M matches the side of $\mathbb{T}$ to which the sector points. For $f = 1 - z$ in H^2 we find $\lambda_M = \log(M|\zeta - 1|) > 0$, so all directions are outward. For the weight (2), the residual of degree $M = M_m$ only sees the last block, where ω is geometric with ratio $e^{-\varepsilon}$, and we find $|B_M| \approx M\varepsilon^2/|1-\zeta|$. This is small for $\varepsilon = M^{-3/4}$ and large for $\varepsilon = M^{-1/4}$, which gives outward and inward zeros, respectively.

For Theorem 3 we use the closed formula of Orive, Sánchez-Lara and Seco [8, Theorem 4.2]: the product $\varphi_n f_\theta f_\theta^*$ has a low part L of degree at most two and a high part $z^{n+3}K$. In $\mathbb{D}$ the high part is exponentially small, so the spurious zeros are essentially the zeros of L. We show that $-(n+3)L$ is close to an explicit quadratic whose coefficients depend on the phase $(n+4)\theta$. As θ moves by $2\pi/(n+4)$ this phase makes a full turn, and the quadratic passes through a double root. By a nested interval argument we find one θ for which the discriminant is at most $2(n+3)^{-8}$ for infinitely many n, so that two spurious zeros sit within $o(1/n)$ of each other. The heuristic behind the conjecture [8, p. 20] assumes that the other zeros do not approach a given spurious zero, and this is exactly what fails here. In the scaled variable $u = n(z - \eta_j)$ the spurious zeros solve $u^2 \approx 0$, while the partner equation becomes $u^2 - 2\xi u \approx 0$.

Organization. The present paper is organized into five additional sections. In Section 2 we describe the residuals for $f = 1 - z$ and $f = (1-z)^2$ and prove a lemma which locates the zeros near ζ, and we prove Theorem 1. In Section 3 we prove Theorem 2. Section 4 contains the proof of Theorem 3, and Section 5 the proof of Theorem 4. Section 6 contains some concluding remarks, among them the regions and rays in the question of Bénéteau and Centner [2, Open Question 2] and the open part of [8, Conjecture 1].

2. The residual and zeros near the circle

We first describe the residual $R = 1 - q_n f$ for the two functions f which we use. Since $q_n f$ is the orthogonal projection of 1 onto $f\mathcal{P}_n$, the residual is characterized by

$$R - 1 \in f\mathcal{P}_n \qquad \text{and} \qquad \langle R, z^j f\rangle_\omega = 0, \quad 0 \le j \le n. \tag{5}$$

Lemma 1. *Suppose that ω satisfies (1), that $\nu \in \{1, 2\}$, that $f(z) = (1-z)^\nu$, and that $n \ge 0$. Let $M = n + \nu$. Then*

$$1 - q_n(z)f(z) = R(z) = \sum_{k=0}^{M} \frac{u_k}{\omega_k} z^k,$$

where $u_k = u_0$ is constant if $\nu = 1$, and $u_k = a + bk$ with real a, b if $\nu = 2$. The constants are determined by $R(1) = 1$ if $\nu = 1$, and by $R(1) = 1$ and $R'(1) = 0$ if $\nu = 2$. The zeros of q_n in $\mathbb{C} \setminus \{1\}$ are the solutions $z \ne 1$ of $R(z) = 1$.

Proof. Write $R(z) = \sum_{k=0}^{M} r_k z^k$ and $u_k = r_k \omega_k$. Since f has real coefficients, $\langle R, z^j f\rangle_\omega$ equals $u_j - u_{j+1}$ if $\nu = 1$ and $u_j - 2u_{j+1} + u_{j+2}$ if $\nu = 2$. By (5) these differences vanish for $0 \le j \le n$, so $(u_k)_{k=0}^{M}$ is constant if $\nu = 1$ and affine in k if $\nu = 2$. A polynomial of degree at most $M = n + \nu$ belongs to $1 + f\mathcal{P}_n$ if and only if $R - 1$ vanishes to order ν at 1. This gives the conditions on $R(1)$ and $R'(1)$. They determine the constants, since q_n is unique. Finally, $q_n f = 1 - R$ and $f(z) \ne 0$ for $z \ne 1$. □

When $\omega_k = 1$ and $\nu = 1$ we get $R(z) = (1 + z + \cdots + z^{n+1})/(n+2)$, and $q_n = (1 - R)/(1 - z)$ is the formula of [2, Example 2.2], which goes back to Bénéteau, Condori, Liaw, Seco and Sola [3].

The next lemma turns the heuristic in the introduction into a statement. We write $z = \zeta e^{s/M}$ and assume that, in the variable s, the equation for the zeros has the form $B_M e^s(1 + \varrho_M(s)) = 1$ with ϱ_M small.

Lemma 2. *Let $\zeta \in \mathbb{T}$ and let Λ be an infinite set of positive integers. For $M \in \Lambda$, let λ_M be real and B_M complex with $|B_M| = e^{-\lambda_M}$, and let*

$$G_M(s) = B_M e^s(1 + \varrho_M(s)) - 1,$$

where ϱ_M is analytic in $\{|s| < (\log M)^2\}$. Suppose that $|\lambda_M| \to \infty$, that $|\lambda_M| \le C \log M$ for some constant C, and that $\sup_{|s|<(\log M)^2} |\varrho_M(s)| \to 0$ as $M \to \infty$ in Λ. Suppose that $\lambda_M \cos\theta_0 > 0$ for every $M \in \Lambda$. Then for every $\delta, \rho > 0$ and every sufficiently large $M \in \Lambda$, there is a zero s_M of G_M with $\zeta e^{s_M/M} \in S(\zeta; \theta_0, \delta, \rho)$.

Proof. Put $t = \tan\theta_0$, which is defined since $\cos\theta_0 \ne 0$. Write $1/B_M = e^{\lambda_M + i\beta_M}$ with β_M real. The solutions of $B_M e^s = 1$ are $\lambda_M + i(\beta_M + 2\pi k)$, $k \in \mathbb{Z}$, and we pick the one closest to the point $\lambda_M(1 + it)$. That is, we choose k with $|\beta_M + 2\pi k - \lambda_M t| \le \pi$ and set $\sigma_M = \lambda_M + i(\beta_M + 2\pi k)$. Then $B_M e^{\sigma_M} = 1$ and

$$|\sigma_M - \lambda_M(1 + it)| \le \pi. \tag{6}$$

Since $|\lambda_M(1+it)| = |\lambda_M|/|\cos\theta_0|$, we have $|\sigma_M| \le C' \log M$, and so the disc $|s-\sigma_M| \le 1/2$ lies in $\{|s| < (\log M)^2\}$ for large M.

On the circle $|s-\sigma_M| = 1/2$ we have $|B_M e^s - 1| = |e^{s-\sigma_M} - 1| \ge c_0$, where

$$c_0 = \min_{|v|=1/2} |e^v - 1| \ge \frac{1}{2} - \left(e^{1/2} - 1 - 1/2\right) > \frac{1}{3},$$

while $|B_M e^s \varrho_M(s)| \le e^{1/2} \sup|\varrho_M| < 1/3$ for large M. By Rouché's theorem, G_M has exactly one zero s_M in $|s-\sigma_M| < 1/2$, because $B_M e^s - 1 = e^{s-\sigma_M} - 1$ has exactly one.

It remains to locate $z_M = \zeta e^{s_M/M}$. Write $z_M = \zeta(1+w_M)$. Since $|e^v - 1 - v| \le |v|^2 e^{|v|}$, we have

$$w_M = e^{s_M/M} - 1 = \frac{s_M}{M}(1+\tau_M), \qquad |\tau_M| \le \frac{|s_M|}{M} e^{|s_M|/M} \to 0.$$

Hence $|w_M| \le 2|s_M|/M = O((\log M)/M)$, and $\arg w_M - \arg s_M \to 0$ modulo 2π. By (6), $|s_M - \lambda_M(1+it)| \le \pi + 1/2$, while $|\lambda_M(1+it)| \to \infty$. So $s_M \ne 0$ for large M, and $\arg s_M - \arg(\lambda_M(1+it)) \to 0$ modulo 2π. If $\cos\theta_0 > 0$, then $\lambda_M > 0$ and $\arg(\lambda_M(1+it)) = \arg(1+it) = \theta_0$ modulo 2π. If $\cos\theta_0 < 0$, then $\lambda_M < 0$ and $\arg(\lambda_M(1+it)) = \arg(1+it) + \pi = \theta_0$ modulo 2π. In both cases $\arg w_M \to \theta_0$ and $|w_M| \to 0$, so $z_M \in S(\zeta; \theta_0, \delta, \rho)$ for large M. □

The proof shows more: near ζ, the zeros of G_M are within $1/2$ of the points $\lambda_M + i(\beta_M + 2\pi k)$ with $|k| = O(\log M)$. This is the comb described in the introduction.

Proof of Theorem 1. (a) This is [5, Corollary 4.3 (ii)], which applies to every nondecreasing weight. We recall the argument. Let $q_n(z_0) = 0$, write $q_n = (z - z_0)p$ and put $g = pf$. Then g and zg lie in $f\mathcal{P}_n$, and $g \ne 0$. Since $1 - (z - z_0)g$ is orthogonal to $f\mathcal{P}_n$, it is orthogonal to zg. As $\langle 1, zg\rangle_\omega = 0$, this gives $\|zg\|_\omega^2 = z_0\langle g, zg\rangle_\omega$. By the Cauchy–Schwarz inequality, $|z_0| \ge \|zg\|_\omega / \|g\|_\omega \ge 1$, where the last inequality holds since ω is nondecreasing. Equality in the Cauchy–Schwarz inequality would make zg a multiple of g, which is impossible. Hence $|z_0| > 1$. If $\cos\theta \le -c < 0$ for $|\theta - \theta_0| \le \delta$, then $|1 + re^{i\theta}|^2 = 1 + 2r\cos\theta + r^2 < 1$ for $0 < r < 2c$, so $S(\zeta; \theta_0, \delta, \rho) \subset \mathbb{D}$ when $\rho \le 2c$.

(b) Let $M = n + 2$, which is one more than the degree $n+1$ of R in Lemma 1. By Lemma 1, $R(z) = (1 - z^M)/(M(1-z))$, so $q_n(z) = 0$ with $z \ne 1$ if and only if $z^M = 1 + M(z-1)$. For $z = \zeta e^{s/M}$ with $|s| < (\log M)^2$ we write

$$1 + M(z-1) = M(\zeta - 1)\big(1 + \gamma_M(s)\big), \qquad \gamma_M(s) = \frac{M(z-\zeta) + 1}{M(\zeta - 1)}.$$

Since $|z - \zeta| \le (|s|/M)e^{|s|/M} \le 2(\log M)^2/M$ for large M, we have $|\gamma_M(s)| \le (2(\log M)^2 + 1)/(M|\zeta - 1|) \to 0$. Hence the zeros of q_n near ζ are the zeros of

$$G_M(s) = B_M e^s\big(1 + \varrho_M(s)\big) - 1, \qquad B_M = \frac{\zeta^M}{M(\zeta - 1)}, \qquad \varrho_M = \frac{1}{1+\gamma_M} - 1,$$

and $\lambda_M = -\log|B_M| = \log(M|\zeta - 1|)$ is positive and of size $\log M$ for large M. Lemma 2 applies for every θ_0 with $\cos\theta_0 > 0$ and gives a zero z_M of $R - 1$ in $S(\zeta; \theta_0, \delta, \rho)$ for every large M. Since $z_M \to \zeta \neq 1$, z_M is a zero of q_{M-2} by Lemma 1. □

In particular, the zeros of q_n near $\zeta \neq 1$ lie at distance about $(\log n)/n$ outside $\mathbb{T}$, with spacing about $2\pi/n$. This is consistent with [5, Theorem 6.1].

3. A weight without zero-free sectors

In this section ω is the weight (2) and $f(z) = (1-z)^2$. We begin with cyclicity. Since $\omega_k \leq 1$, we have $\|h\|_\omega \leq \|h\|_{H^2}$ for every h. The polynomial $(1-z)^2$ has no zeros in $\mathbb{D}$, so it is outer and hence cyclic in H^2 by Beurling's theorem [6, Chapter 7]. Hence there are polynomials p_j with $\|p_j f - 1\|_\omega \leq \|p_j f - 1\|_{H^2} \to 0$, and f is cyclic in H^2_ω.

Before the estimates, let us explain why we use $(1-z)^2$ rather than $1 - z$. By Lemma 1, the residual for $f = 1 - z$ is $R(z) = \sum_k z^k/\omega_k$ divided by $\sum_k 1/\omega_k$. Reversing the order of the coefficients, the equation $R(z) = 1$ near ζ involves the normalized sum $\sum_j v_j x^j / \sum_j v_j$ with $x = 1/z$ and positive v_j. On a geometric block $v_j = e^{-\varepsilon j}$ this is about $(1 - e^{-\varepsilon})/(1 - \overline{\zeta})$, which is small. As in the proof of Theorem 1, this puts the zeros near ζ outside the disc. For $f = (1-z)^2$ the coefficients $a + bk$ of u_k must satisfy two conditions. They nearly cancel at $x = 1$, where $R(1) = 1$ is prescribed, but not at $x = \overline{\zeta}$, and this produces the large factor $M\varepsilon^2$ in Lemma 3 below.

Fix $\zeta \in \mathbb{T} \setminus \{1\}$. In the rest of this section m is large, and we write

$$M = M_m, \qquad \varepsilon = \varepsilon_m, \qquad L = M_m - M_{m-1} = 3M/4, \qquad q = e^{-\varepsilon},$$

and $n = M - 2$. Then $M^{-3/4} \leq \varepsilon \leq M^{-1/4}$ and $\varepsilon L \geq 3M^{1/4}/4$. Put $v_j = \omega_M/\omega_{M-j}$ for $0 \leq j \leq M$. By (2), and since ω is nonincreasing,

$$v_j = q^j \quad (0 \leq j \leq L), \qquad 0 < v_j \leq q^L \quad (L < j \leq M). \tag{7}$$

So the residual of degree M sees the last block of the weight, where ω is geometric, and everything earlier contributes at most $q^L \leq e^{-3M^{1/4}/4}$. Substituting $k = M - j$ in Lemma 1, we find that $R(z) = z^M P(1/z)$, where

$$P(x) = \sum_{j=0}^{M} (A + Bj) v_j x^j, \qquad A = \frac{a + bM}{\omega_M}, \qquad B = -\frac{b}{\omega_M}.$$

The conditions $R(1) = 1$ and $R'(1) = 0$ become

$$\sum_{j=0}^{M} (A + Bj) v_j = 1, \qquad \sum_{j=0}^{M} j(A + Bj) v_j = M, \tag{8}$$

since $R'(1) = \sum_k k r_k = MR(1) - \sum_j j(A + Bj)v_j$. In the estimates below the constants depend only on ζ, and C denotes an absolute constant whose value may change from line to line.

Lemma 3. *Let $P_0 = -M\varepsilon^2/(1-\overline{\zeta})$. Then*

$$\lim_{m\to\infty} \sup\left\{\left|\frac{P(x)}{P_0} - 1\right| :\ x = \overline{\zeta}e^{-s/M},\ |s| < (\log M)^2\right\} = 0.$$

Proof. We compare P with the polynomial we would get if the weight were geometric for all $k \le M$, and then evaluate the geometric model in closed form.

The geometric model. Let $g_i = \sum_{j\ge 0} j^i q^j$ for $i = 0, 1, 2$, so that $g_0 = 1/(1-q)$, $g_1 = q/(1-q)^2$ and $g_2 = q(1+q)/(1-q)^3$. A direct computation shows that

$$A_g = (1-q)\big(1+q-M(1-q)\big), \qquad B_g = \frac{(1-q)^2\big(M(1-q)-q\big)}{q}$$

solve $A_g g_0 + B_g g_1 = 1$ and $A_g g_1 + B_g g_2 = M$, the analogue of (8) with $v_j = q^j$ for all $j \ge 0$. For $|qx| < 1$ put

$$P_g(x) = \sum_{j=0}^{\infty} (A_g + B_g j) q^j x^j = \frac{A_g}{1-qx} + \frac{B_g qx}{(1-qx)^2}.$$

Comparison of the coefficients. Let $n_i = \sum_{j=0}^{M} j^i v_j$. By (7), and since $1 - q^{1/2} \ge \varepsilon/4$,

$$|n_i - g_i| \le (M+1)^3 q^L + \sum_{j>L} j^2 q^j \le (M+1)^3 q^L + q^{L/2} \sum_{j\ge 0} j^2 q^{j/2} \le M^C e^{-\varepsilon L/2}$$

for $i = 0, 1, 2$. By Cramer's rule, $A = (n_2 - M n_1)/\Delta$ and $B = (M n_0 - n_1)/\Delta$ with $\Delta = n_0 n_2 - n_1^2$, and similarly for A_g and B_g with $\Delta_g = g_0 g_2 - g_1^2 = q/(1-q)^4 \ge e^{-1}$. Each of n_i and g_i is at most M^C, since $1/\varepsilon \le M$. Hence $|\Delta - \Delta_g| \le M^C e^{-\varepsilon L/2}$, and therefore $|A - A_g| + |B - B_g| \le M^C e^{-\varepsilon L/2}$ for large m.

Comparison of P and P_g. Let $x = \overline{\zeta}e^{-s/M}$ with $|s| < (\log M)^2$. Then $|x|^j \le e^{(\log M)^2}$ for $0 \le j \le M$, and $|qx| \le e^{-\varepsilon + (\log M)^2/M} \le e^{-\varepsilon/2}$ for large m. Moreover $|A_g| + |B_g| \le 3eM$. We write

$$P(x) - P_g(x) = \sum_{j=0}^{M} \big((A - A_g) + (B - B_g)j\big) v_j x^j + \sum_{L<j\le M} (A_g + B_g j)(v_j - q^j) x^j - \sum_{j>M} (A_g + B_g j)(qx)^j.$$

Using $v_j \le 1$, (7) and $|qx| \le e^{-\varepsilon/2}$, we find that each of the three sums is at most $M^C e^{(\log M)^2} e^{-\varepsilon L/2}$. Since $\varepsilon L \ge 3M^{1/4}/4$ and $|P_0| \ge M^{-1/2}/2$, this is $o(|P_0|)$, uniformly in s.

Evaluation of P_g. Put $y = qx$. Since $|x - \overline{\zeta}| \le 2|s|/M$ and $1 - q \le \varepsilon$, we have $|y - \overline{\zeta}| \le 2\varepsilon$ for large m, and hence $1/(1-y) = (1 + O(\varepsilon))/(1-\overline{\zeta})$. Since $1 - q = \varepsilon(1 + O(\varepsilon))$, we get

$$A_g = -M(1-q)^2 + (1-q^2) = -M\varepsilon^2\Big(1 + O(\varepsilon) + O\Big(\frac{1}{M\varepsilon}\Big)\Big), \qquad B_g = O(M\varepsilon^3).$$

Thus the first term of $P_g(x)$ is $P_0(1+O(\varepsilon)+O(1/(M\varepsilon)))$, and the second is $O(M\varepsilon^3) = P_0 \cdot O(\varepsilon)$. Since $\varepsilon \le M^{-1/4}$ and $M\varepsilon \ge M^{1/4}$, we conclude that $P_g(x)/P_0 \to 1$ uniformly, and together with the previous step this proves the lemma. □

At $x = 1$ the two terms of P_g are of size $M\varepsilon$ and cancel to give $P_g(1) = 1$. At $x = \overline{\zeta}$ there is no such cancellation, and $|P_0| = M\varepsilon^2/|1-\zeta|$ is large or small according to whether ε is larger or smaller than $M^{-1/2}$.

Proof of Theorem 2. Cyclicity was shown at the beginning of this section. Let Λ be the set of M_m with m odd if $\cos\theta_0 > 0$ and with m even if $\cos\theta_0 < 0$. For $z = \zeta e^{s/M}$ we have $z^M = \zeta^M e^s$ and $1/z = \overline{\zeta} e^{-s/M}$, and so

$$R(\zeta e^{s/M}) - 1 = B_M e^s (1 + \varrho_M(s)) - 1,$$

where

$$B_M = \zeta^M P_0, \qquad \varrho_M(s) = \frac{P(\overline{\zeta} e^{-s/M})}{P_0} - 1.$$

By Lemma 3, $\sup_{|s|<(\log M)^2} |\varrho_M(s)| \to 0$. Moreover

$$\lambda_M = -\log|B_M| = \log\frac{|1-\zeta|}{M\varepsilon^2} = \log|1-\zeta| \pm \frac{1}{2}\log M,$$

with the sign + when m is odd ($\varepsilon = M^{-3/4}$) and − when m is even ($\varepsilon = M^{-1/4}$). Hence $|\lambda_M| \to \infty$, $|\lambda_M| \le \log M$, and $\lambda_M \cos\theta_0 > 0$ for large $M \in \Lambda$. By Lemma 2, for every large $M \in \Lambda$ there is a zero z_M of $R - 1$ in $S(\zeta; \theta_0, \delta, \rho)$. Since $z_M \to \zeta \ne 1$, z_M is a zero of q_{M-2} by Lemma 1. □

4. Colliding spurious zeros

In this section we work in H^2, fix θ in a compact interval $I \subset (0, \pi)$ and write $a = e^{i\theta}$, $f = f_\theta$ as in (3), and $N = n+3$. Let $w = |f|^2$ on $\mathbb{T}$, and let φ_n be the monic orthogonal polynomial of degree n for $w\,\mathrm{d}m$. The zeros $1, a, \overline{a}$ of f are called the *nodes*. Let $f^*(z) = z^3\overline{f(1/\overline{z})}$. Since f has real coefficients and $z^2 - 2z\cos\theta + 1$ is self-reciprocal, $f^* = -f$. On $\mathbb{T}$ we have $ff^* = z^3|f|^2$. The leading coefficient of f is -1.

The low and high parts. The following decomposition is [8, Theorem 4.2 and (25)], up to normalization; we include the short proof. For $3 \le k \le n+2$, the coefficient of z^k in $ff^*\varphi_n$ is

$$\int_{\mathbb{T}} ff^*\varphi_n \overline{z}^k \,\mathrm{d}m = \int_{\mathbb{T}} \varphi_n \,\overline{z^{k-3}}\, w \,\mathrm{d}m = 0$$

by orthogonality. Since $ff^*\varphi_n = -f^2\varphi_n$ has degree $N+3$ and leading coefficient -1, we may write

$$-f^2\varphi_n = L + z^N K, \qquad \deg L \le 2, \qquad \deg K = 3, \tag{9}$$

where K has leading coefficient -1. The polynomials L and K have real coefficients, because w is symmetric under conjugation, and they depend

continuously on θ, because the coefficients of φ_n solve a linear system with a positive definite Gram matrix depending continuously on θ.

Since φ_n has no zeros on $\mathbb{T}$, the zeros of φ_n in $\mathbb{D}$ are the zeros of $L + z^N K$ in $\mathbb{D}$. For large n the term $z^N K$ is exponentially small on compact subsets of $\mathbb{D}$, so the spurious zeros are essentially the zeros of the quadratic L. The next lemma computes L to leading order. The key information is that $-f^2\varphi_n$ has a double zero at each node.

Lemma 4. *For $\psi \in \mathbb{R}$ let*

$$Q_{\theta,\psi}(z) = (1+2\cos\psi)z^2 - 2\big(\cos\theta+\cos\psi+\cos(\psi-\theta)\big)z + 1 + 2\cos(\psi-\theta). \tag{10}$$

Let $Q_n = -NL$. Then, in the norm given by the sum of the absolute values of the coefficients and uniformly for $\theta \in I$,

$$K = f + O(1/N), \qquad Q_n = Q_{\theta,(N+1)\theta} + O(1/N).$$

Proof. Since K and f both have degree three and leading coefficient -1, we may write $K = f + R$ with $\deg R \le 2$. Let b be a node. Since $-f^2\varphi_n$ vanishes to second order at b, (9) gives

$$L(b) + b^N K(b) = 0, \qquad L'(b) + Nb^{N-1}K(b) + b^N K'(b) = 0.$$

Since $f(b) = 0$, we have $K(b) = R(b)$ and $K'(b) = f'(b) + R'(b)$. Inserting the first identity into the second and multiplying by b, we obtain

$$R(b) = -b^{-N}L(b), \qquad NL(b) = bL'(b) + b^{N+1}\big(f'(b) + R'(b)\big). \tag{11}$$

We now read (11) as a linear equation for L in the three-dimensional space $\mathcal{P}_2$. For complex numbers (y_b) indexed by the nodes, let $\mathcal{I}[y] \in \mathcal{P}_2$ be the Lagrange interpolant with $\mathcal{I}[y](b) = y_b$. The nodes lie on $\mathbb{T}$ and are separated by a positive distance which depends only on I, so $\|\mathcal{I}[y]\| \le C \max_b |y_b|$ with C depending only on I. Every polynomial in $\mathcal{P}_2$ is the interpolant of its values at the nodes. Hence the first identity in (11) says that $R = -T_N L$, where $T_N L = \mathcal{I}[b^{-N}L(b)]$, and the second says that

$$NL = \mathcal{B}_N L + V_N, \qquad V_N = \mathcal{I}\big[b^{N+1}f'(b)\big],$$
$$\mathcal{B}_N L = \mathcal{I}\big[bL'(b) - b^{N+1}(T_N L)'(b)\big].$$

Since $|b| = 1$, the operators T_N and $\mathcal{B}_N$ on $\mathcal{P}_2$ are bounded uniformly in N and $\theta \in I$, and $\|V_N\| \le C$. Hence, for large N,

$$L = \frac{1}{N}\Big(1 - \frac{\mathcal{B}_N}{N}\Big)^{-1} V_N = \frac{V_N}{N} + O(N^{-2})$$

by the Neumann series, and $R = -T_N L = O(1/N)$. This proves $K = f + O(1/N)$ and $Q_n = -V_N + O(1/N)$.

It remains to compute V_N. The Lagrange basis polynomial for the node b is $f(z)/((z-b)f'(b))$, so

$$\begin{aligned} V_N(z) &= f(z)\sum_b \frac{b^{N+1}}{z-b} \\ &= -\Big((z-a)(z-\overline{a}) + a^{N+1}(z-1)(z-\overline{a}) + \overline{a}^{N+1}(z-1)(z-a)\Big), \end{aligned}$$

where we used $f(z) = -(z-1)(z-a)(z-\overline{a})$. With $\psi = (N+1)\theta$, the last two terms are $e^{i\psi}(z^2 - (1+\overline{a})z + \overline{a})$ and its coefficientwise conjugate. Expanding, we find that $-V_N = Q_{\theta,\psi}$. □

Lemma 4 agrees with the interior asymptotics of Martínez-Finkelshtein, McLaughlin and Saff [7, Theorem 1 (i)], where $n\varphi_n$ is asymptotic to a fixed function times $\sum_k \beta_k \vartheta_k a_k^{n+1}/(a_k - z)$. For us the relevant feature is that the quadratic depends on n only through the phase $\psi = (N+1)\theta$ modulo 2π.

Windows with a nearly double root. For a real quadratic $Q(z) = \mathsf{A}z^2 + \mathsf{B}z + \mathsf{C}$ with $\mathsf{A} \neq 0$, let $\Delta(Q) = \mathsf{B}^2 - 4\mathsf{AC}$ be its discriminant and $\eta(Q) = -\mathsf{B}/(2\mathsf{A})$ the midpoint of its zeros. At $\theta = \pi/2$, with $c = \cos\psi$ and $s = \sin\psi$, (10) becomes

$$Q_{\pi/2,\psi}(z) = (1+2c)z^2 - 2(c+s)z + 1 + 2s, \qquad \Delta(Q_{\pi/2,\psi}) = -8(c+s+cs).$$

Let $y = \sqrt{2} - 1$, and choose $\psi_0 \in (-\pi/2, 0)$ with

$$c_0 = \cos\psi_0 = \frac{y + \sqrt{2-y^2}}{2}, \qquad s_0 = \sin\psi_0 = \frac{y - \sqrt{2-y^2}}{2}.$$

These satisfy $c_0^2 + s_0^2 = 1$ and $c_0 + s_0 = y$, $c_0 s_0 = (y^2-1)/2$, so that $c_0 + s_0 + c_0 s_0 = (y^2 + 2y - 1)/2 = 0$. Hence $Q_{\pi/2,\psi_0}$ has the double zero

$$\xi_0 = \frac{c_0 + s_0}{1 + 2c_0} = \frac{y}{1+2c_0} = 0.1497\ldots$$

and positive leading coefficient $1 + 2c_0$. Moreover

$$\partial_\psi \Delta(Q_{\pi/2,\psi})\big|_{\psi=\psi_0} = -8(c_0 - s_0)(1 + c_0 + s_0) < 0,$$

so the discriminant changes sign transversally at ψ_0. Fix numbers $0 < \alpha < \xi_0 < \beta < 1/4$ and $0 < \mathsf{A}_- < 1 + 2c_0 < \mathsf{A}_+$. By the implicit function theorem and continuity, there are a compact interval I around $\pi/2$, a continuous function $\psi(\theta)$ on I with $\psi(\pi/2) = \psi_0$ and $\Delta(Q_{\theta,\psi(\theta)}) = 0$, and numbers $\varepsilon > 0$ and $\kappa > 0$ such that the following holds whenever θ and θ' lie in I and $|\psi - \psi(\theta')| \le \varepsilon$: the leading coefficient of $Q_{\theta,\psi}$ lies in $[\mathsf{A}_- + \kappa, \mathsf{A}_+ - \kappa]$, $\eta(Q_{\theta,\psi})$ lies in $[\alpha + \kappa, \beta - \kappa]$, and

$$\Delta\big(Q_{\theta,\psi(\theta')-\varepsilon}\big) \ge \kappa, \qquad \Delta\big(Q_{\theta,\psi(\theta')+\varepsilon}\big) \le -\kappa. \tag{12}$$

We use this I in Lemma 4 from now on. In particular, the leading coefficient of Q_n is positive whenever the phase is in such a window, and we write $Q_n(z) = \mathsf{A}_n z^2 + \mathsf{B}_n z + \mathsf{C}_n$.

The next lemma is where the rapid rotation of the phase is used: as θ moves by $2\pi/(N+1)$, the phase $(N+1)\theta$ makes a full turn, while the coefficients depending on θ itself barely change.

Lemma 5. *Suppose that J is a nonempty open interval contained in I and that n_0 is an integer. Then there are $n > n_0$ and a nonempty open interval $U \subset J$ such that for every $\theta \in U$*

$$\mathsf{A}_- < \mathsf{A}_n < \mathsf{A}_+, \qquad \alpha < \eta(Q_n) < \beta, \qquad N^{-8} < \Delta(Q_n) < 2N^{-8}. \tag{13}$$

Proof. Fix $\theta_* \in J$. For large N, let k_N be an integer nearest to $((N+1)\theta_* - \psi(\theta_*))/(2\pi)$, and put

$$\theta_N^{\pm} = \frac{2\pi k_N + \psi(\theta_*) \pm \varepsilon}{N+1}.$$

Then $|\theta_N^{\pm} - \theta_*| \le (\pi+\varepsilon)/(N+1)$, so the closed interval between θ_N^- and θ_N^+ lies in J for large N. For θ in this interval, $(N+1)\theta - 2\pi k_N$ lies within ε of $\psi(\theta_*)$, and at the endpoints it equals $\psi(\theta_*) \pm \varepsilon$. Since Δ, the leading coefficient and η are continuous functions of the coefficients, Lemma 4 and the choice of I, ε and κ show that, for large N, the first two conditions in (13) hold on the whole interval, while $\Delta(Q_n) \ge \kappa/2$ at θ_N^- and $\Delta(Q_n) \le -\kappa/2$ at θ_N^+ by (12). Since $\theta \mapsto \Delta(Q_n)$ is continuous, it takes the value $3N^{-8}/2$ at some θ' between θ_N^- and θ_N^+. All three conditions in (13) are open, so they hold on an open interval $U \ni \theta'$. Finally we take N so large that $n = N - 3 > n_0$. □

For a single n, Lemma 5 produces a parameter θ depending on n. A weight depending on n would not give a counterexample, so in the proof of Theorem 3 we freeze θ by a nested interval argument.

The electrostatic partner. We recall the definition from [8, Section 2]. The weight $W_n(z) = z^{-n}\prod_j z^{-1}(z-a_j)^2$, with a_j running over the nodes, is semiclassical: by [8, (22)],

$$\frac{W_n'}{W_n} = \frac{\mathcal{B}_n}{\mathcal{A}}, \qquad \mathcal{A}(z) = z\prod_j (z-a_j) = -zf(z),$$

so $\mathcal{B}_n = \mathcal{A}(-N/z + 2f'/f)$. The second-kind function is

$$\widehat{\varphi}_n(z) = \int_{\mathbb{T}} \frac{\varphi_n(t) W_n(t)}{\lambda(t)(t-z)}\,\mathrm{d}t,$$

where λ is the function from [8, Lemma 4.1]; since our exponents are integers, λ is constant, and W_n/λ is a constant multiple of $t^{-n}w(t) = -t^{-N}f(t)^2$ on $\mathbb{T}$. The electrostatic partner is [8, (7) and (26)]

$$S_n = c\,(\mathcal{A}(\varphi_n\widehat{\varphi}_n' - \varphi_n'\widehat{\varphi}_n) - \mathcal{B}_n\varphi_n\widehat{\varphi}_n),$$

with c chosen so that S_n is monic. Multiplying $\widehat{\varphi}_n$ by a constant does not change S_n. Since S_n is a polynomial, it is determined by its values in $\{|z| > 1\}$, and there we may use the closed formula for $\widehat{\varphi}_n$ in the next proof.

Lemma 6. *The partner S_n is a constant multiple of*

$$\frac{z(KL' - K'L) - NKL}{f}, \tag{14}$$

which is a nonzero polynomial of degree at most two. Let $0 < r < 1$. For large n, the zeros of S_n in $\{|z| \le r\}$ are the zeros there of

$$H_n(z) = Q_n(z) - \frac{z}{N}Q_n'(z) + \frac{z}{N}\frac{K'(z)}{K(z)}Q_n(z). \tag{15}$$

Proof. Up to a constant factor, $\widehat{\varphi}_n(z) = \int_{\mathbb{T}} \varphi_n(t)t^{-N}(-f(t)^2)\,\mathrm{d}t/(t-z)$. For $|z| > 1$ this equals $-2\pi i z^{-N}L(z)$ [8, Corollary 4.1]. Indeed, by (9) the integrand is $(t^{-N}L(t) + K(t))/(t-z)$; the polynomial K contributes nothing, and $t^{-N}L(t)/(t-z)$ is analytic in $|t| > 1$ except at $t = z$ and decays like $|t|^{-2}$, so its integral is minus the residue at z times $2\pi i$.

We now compute S_n with $\varphi_n = P/D$, where $P = L + z^N K$ and $D = -f^2$, and with $\widehat{\varphi}_n$ replaced by $Y = z^{-N}L$. First,

$$\varphi_n Y' - \varphi_n' Y = \frac{PY' - P'Y}{D} + \frac{D'}{D^2}PY, \qquad \mathcal{B}_n\varphi_n Y = \mathcal{A}\Big(\frac{D'}{D} - \frac{N}{z}\Big)\frac{PY}{D},$$

since $D'/D = 2f'/f$. In the difference the terms with D' cancel, and we obtain

$$\mathcal{A}(\varphi_n Y' - \varphi_n' Y) - \mathcal{B}_n\varphi_n Y = \frac{\mathcal{A}}{D}\Big(PY' - P'Y + \frac{N}{z}PY\Big).$$

Since $Y' = z^{-N}L' - Nz^{-N-1}L$, the bracket equals $z^{-N}(PL' - P'L)$. Inserting $P = L + z^N K$, we get

$$z^{-N}(PL' - P'L) = KL' - K'L - \frac{N}{z}KL.$$

Since $\mathcal{A}/(zD) = -zf/(-zf^2) = 1/f$, this proves (14).

At a node b, (9) gives $L(b) = -b^N K(b)$ and $L'(b) = -Nb^{N-1}K(b) - b^N K'(b)$. Inserting these, we find that the numerator of (14) vanishes at b. Hence (14) is a polynomial. The numerator has degree at most five, so the quotient has degree at most two. If the numerator vanished identically, then the derivative of $L/(z^N K)$, which is $z^{N-1}(z(KL' - K'L) - NKL)/(z^N K)^2$, would vanish, so $L = \gamma z^N K$ for a constant γ. As $\deg L \le 2 < N$, this forces $L = 0$, and then $-f^2\varphi_n = z^N K$, so that f^2 divides K, which is impossible since $\deg K = 3$.

Finally, let $|z| \le r$. Since $K = f + O(1/N)$ and f has no zeros in $\overline{\mathbb{D}}$ apart from the nodes on $\mathbb{T}$, K has no zeros in $\{|z| \le r\}$ for large n, and K'/K is bounded there uniformly in n and $\theta \in I$. Substituting $L = -Q_n/N$ and $L' = -Q_n'/N$, we find that the numerator of (14) equals KH_n. Since f and K have no zeros in $\{|z| \le r\}$, this proves the last assertion. □

Proof of Theorem 3. Let J_0 be a compact interval with nonempty interior in the interior of I, and let $n_0 = 0$. For $j \geq 1$ we apply Lemma 5 to the interior of J_{j-1} and n_{j-1}. This gives $n_j > n_{j-1}$ and an open interval U_j in the interior of J_{j-1} on which (13) holds with $n = n_j$. Let $J_j \subset U_j$ be a compact interval with nonempty interior and length at most 2^{-j}. The intervals J_j are nested, so they have exactly one common point θ. Since $\theta \in U_j$ for every j, (13) holds with $n = n_j$ for every j. From now on θ is fixed.

Write $N_j = n_j + 3$, $Q_j = Q_{n_j}$, $\mathsf{A}_j = \mathsf{A}_{n_j}$ and $\eta_j = \eta(Q_j) \in (\alpha, \beta)$. Passing to a subsequence, we may assume that $\eta_j \to \xi \in [\alpha, \beta] \subset (0, 1/4)$. Completing the square,

$$Q_j(z) = \mathsf{A}_j((z - \eta_j)^2 - \varrho_j^2), \qquad \varrho_j^2 = \frac{\Delta(Q_j)}{4\mathsf{A}_j^2} = O(N_j^{-8}).$$

We use the scaled variable $u = N_j(z - \eta_j)$.

The spurious zeros. By (9), the zeros of φ_{n_j} in $\mathbb{D}$ are the zeros of $L + z^N K$, that is, of $Q_j - N_j z^{N_j} K$. Let

$$\begin{aligned} Z_j(u) &= \frac{N_j^2}{\mathsf{A}_j}(Q_j(z) - N_j z^{N_j} K(z)) \\ &= u^2 - N_j^2 \varrho_j^2 - \frac{N_j^3}{\mathsf{A}_j} z^{N_j} K(z), \qquad z = \eta_j + \frac{u}{N_j}. \end{aligned}$$

Fix r_0 with $\beta < r_0 < 1$. For u in a fixed compact set and large j we have $|z| \leq r_0$, so the last term is at most $CN_j^3 r_0^{N_j}$, and $N_j^2 \varrho_j^2 = O(N_j^{-6})$. Hence $Z_j(u) \to u^2$ locally uniformly. By Hurwitz's theorem, for every $\delta > 0$ and large j, Z_j has exactly two zeros in $|u| < \delta$. Call the corresponding zeros of φ_{n_j} $\xi_{j,1}$ and $\xi_{j,2}$. Then $N_j(\xi_{j,k} - \eta_j) \to 0$, and also $n_j(\xi_{j,k} - \eta_j) \to 0$, since $n_j/N_j \to 1$.

Let $1/4 < r < 1$. On $|z| = r$ we have $|Q_j(z)| \geq \mathsf{A}_-((r - \beta)^2 - \varrho_j^2)$, which is bounded below, while $N_j|z^{N_j} K(z)| \leq CN_j r^{N_j} \to 0$. By Rouché's theorem, $Q_j - N_j z^{N_j} K$ has as many zeros in $|z| < r$ as Q_j, namely two. Hence $\xi_{j,1}$ and $\xi_{j,2}$ are the only zeros of φ_{n_j} in $\{|z| \leq r\}$ for large j. In particular, all the other zeros of φ_{n_j} accumulate on $\mathbb{T}$.

The partner zeros. By (15) and $Q_j'(z) = 2\mathsf{A}_j(z - \eta_j)$,

$$\frac{N_j^2}{\mathsf{A}_j} H_{n_j}\Big(\eta_j + \frac{u}{N_j}\Big) = u^2 - N_j^2 \varrho_j^2 - 2\Big(\eta_j + \frac{u}{N_j}\Big)u + \frac{N_j}{\mathsf{A}_j}\, z\, \frac{K'(z)}{K(z)} Q_j(z),$$

where $z = \eta_j + u/N_j$. On a compact set of u, the last term is $O(N_j \cdot N_j^{-2}) = O(1/N_j)$, because $Q_j(z) = O(N_j^{-2})$ there and K'/K is bounded. Hence the left-hand side tends to $u^2 - 2\xi u = u(u - 2\xi)$ locally uniformly. The two limiting zeros 0 and 2ξ are distinct, so by Hurwitz's theorem H_{n_j} has zeros $s_{j,1}$ and $s_{j,2}$ with $N_j(s_{j,1} - \eta_j) \to 0$ and $N_j(s_{j,2} - \eta_j) \to 2\xi$. By Lemma 6 these are zeros of S_{n_j}, which has degree at most two; hence S_{n_j} has degree

two and these are all its zeros. They lie in $\{|z| < r\}$ for large j, since $s_{j,k} \to \xi < 1/4$. This proves (a) and (b).

The pairing. Since $n_j(\xi_{j,k} - \eta_j) \to 0$ and $s_{j,\ell} \to \xi$,

$$n_j\Big(\xi_{j,k} - s_{j,\ell} + \frac{s_{j,\ell}}{n_j}\Big) = n_j(\xi_{j,k} - \eta_j) - n_j(s_{j,\ell} - \eta_j) + s_{j,\ell}$$

tends to $0 - 0 + \xi = \xi$ if $\ell = 1$ and to $0 - 2\xi + \xi = -\xi$ if $\ell = 2$, for $k = 1, 2$. Every bijection σ uses both values of ℓ, so the maximum in (4) tends to ξ. □

5. Weak asymptotics of the zeros of the opas

In this section we prove Theorem 4. Let $W = |F|^2$ on $\mathbb{T}$ and let φ_n be the monic orthogonal polynomials for $W\,\mathrm{d}m$. On $\mathbb{T}$,

$$W(z) = |g(z)|^2 \prod_{k=1}^{m} |z - a_k|^{2\beta_k},$$

and $|g|^2$ is positive on $\mathbb{T}$ and extends to the analytic zero-free function $g(z)\overline{g(1/\overline{z})}$ in an annulus around $\mathbb{T}$. Since $\beta_k > 0 > -1/2$, the results of Martínez-Finkelshtein, McLaughlin and Saff apply. We use two of them.

- (M1) For every compact $K \subset \mathbb{D}$ there is n_K such that φ_n has at most $m - 1$ zeros in K for $n \geq n_K$ [7, Corollary 1].
- (M2) $z^{-n}\varphi_n(z)$ converges uniformly on compact subsets of $\{|z| > 1\}$ to an analytic function [7, Theorem 1 (iii)].

Proof of Theorem 4. Zeros of φ_n. Every zero ζ of φ_n lies in $\mathbb{D}$. Indeed, write $\varphi_n = (z - \zeta)h$ with $\deg h = n - 1$. By orthogonality $\int_{\mathbb{T}}(z - \zeta)|h|^2 W\,\mathrm{d}m = 0$, so

$$\zeta = \frac{\int_{\mathbb{T}} z|h|^2 W\,\mathrm{d}m}{\int_{\mathbb{T}}|h|^2 W\,\mathrm{d}m},$$

a weighted average of the points of $\mathbb{T}$ with respect to a measure which is not a point mass. Let μ_n be the zero counting measure of φ_n divided by n. By (M1), every weak limit point μ of (μ_n) is supported on $\mathbb{T}$. Fix $R > 1$. By (M2), $|z^{-n}\varphi_n(z)|$ is bounded on $|z| = R$ uniformly in n. The coefficient of z^{-k} in $z^{-n}\varphi_n(z)$ is the coefficient of z^{n-k} in φ_n, so by Cauchy's estimate each of the first k coefficients of φ_n below the leading one is bounded uniformly in n. By Newton's identities, the power sums $\sum_{\varphi_n(\zeta)=0}\zeta^k$ are bounded in n for each $k \geq 1$. Hence $\int z^k\,\mathrm{d}\mu = 0$ for $k \geq 1$, and since μ lives on $\mathbb{T}$, also for $k \leq -1$. So $\mu = m$, and $\mu_n \to m$ weakly.

From φ_n to p_n. Let $\varphi_n^*(z) = z^n\overline{\varphi_n(1/\overline{z})}$. For $1 \leq j \leq n$,

$$\langle \varphi_n^*, z^j\rangle_W = \int_{\mathbb{T}} z^{n-j}\overline{\varphi_n} W\,\mathrm{d}m = \overline{\langle \varphi_n, z^{n-j}\rangle_W} = 0,$$

and since $\varphi_n^*(0) = 1$, $\varphi_n^* - 1$ lies in the span of $z, \dots, z^n$, so $\langle \varphi_n^*, 1\rangle_W = \|\varphi_n^*\|_W^2$. The opa p_n is characterized by $\langle p_n F, z^j F\rangle_{H^2} = \langle 1, z^j F\rangle_{H^2}$ for $0 \leq j \leq n$,

that is, $\langle p_n, z^j\rangle_W = \overline{F(0)}$ if $j = 0$ and 0 if $1 \le j \le n$. Hence

$$p_n = \frac{\overline{F(0)}}{\|\varphi_n^*\|_W^2}\,\varphi_n^*,$$

which is [8, (2)], here valid for every n. The zeros of p_n are the points $1/\overline{\zeta}$, where $\zeta \neq 0$ is a zero of φ_n, and $n - \deg p_n$ is the multiplicity of 0 as a zero of φ_n. By (M1) with $K = \{0\}$, $\deg p_n \ge n - m + 1$ for large n.

Let χ be a bounded continuous function on $\mathbb{C}$ and $R > 1$. By (M1) with $K = \{|z| \le 1/R\}$, all but at most $m - 1$ zeros ζ of φ_n satisfy $1/R < |\zeta| < 1$, and then $1 < |1/\overline{\zeta}| < R$ and $|1/\overline{\zeta} - \zeta| < R - 1/R$. Let ν_n be the zero counting measure of p_n divided by $\deg p_n$. Discarding the at most $m - 1$ exceptional zeros, some of which may tend to infinity, and using $\deg p_n = n + O(1)$, we find that

$$\limsup_{n\to\infty} \Big| \int \chi \,\mathrm{d}\nu_n - \int \chi \,\mathrm{d}\mu_n \Big| \le \sup_{(z,z')\in E_R} |\chi(z) - \chi(z')|,$$

where E_R is the set of pairs with $1/R \le |z|, |z'| \le R$ and $|z - z'| \le R - 1/R$. By uniform continuity of χ on $\{1/2 \le |z| \le 2\}$, the right-hand side tends to 0 as $R \to 1^+$. Since $\int \chi \,\mathrm{d}\mu_n \to \int \chi \,\mathrm{d}m$, we conclude that $\nu_n \to m$ weakly. □

6. Concluding remarks

Regions and rays. The second half of [2, Open Question 2] asks for regions or rays avoided by the zeros. For compact subsets of $\mathbb{D}$ the answer is yes in general, by a standard argument which we include for completeness; compare [5, Proposition 6.3].

Proposition 1. *Suppose that ω satisfies (1), that $f \in H^2_\omega$ is cyclic and that $K \subset \mathbb{D}$ is compact. Then q_n has no zeros in K for all sufficiently large n.*

Proof. Since $\omega_k/\omega_{k+1} \to 1$, we have $\omega_k^{1/k} \to 1$, so $k_\omega(z) = \sum_k |z|^{2k}/\omega_k$ is bounded on K. By the Cauchy–Schwarz inequality, $|h(z)|^2 \le \|h\|^2_\omega k_\omega(z)$ for $h \in H^2_\omega$. Cyclicity gives $\|q_n f - 1\|_\omega \to 0$, so $q_n f \to 1$ uniformly on K. In particular f has no zeros in K, and $|q_n| \ge (2\max_K |f|)^{-1}$ on K for large n. □

Rays alone carry little information: each $q_n \not\equiv 0$ has finitely many zeros, so the set of all zeros of all opas is countable, and all but countably many rays from ζ contain no zero at all. For neighbourhoods of rays, which are the sectors, Theorem 2 gives the negative answer.

What Theorem 2 does not decide. Theorem 2 concerns one admissible weight, which answers the question in the generality of (1). This weight is not quasi-integral: the norm of a rotation invariant measure on $\overline{\mathbb{D}}$ is given by moments $h(k) = \int r^{2k} \,\mathrm{d}\nu(r)$, and $\log h$ is convex, while $\log \omega$ alternates between long blocks of slopes $-M_m^{-1/4}$ and $-M_m^{-3/4}$ and stays within no bounded distance of a convex function. So the weight lies outside the class of the Jentzsch-type theorems which motivate the question, and Theorem 2

does not decide the question for quasi-integral weights. In particular, it does not decide it for specific spaces in which opas may vanish inside the disc, such as D_α with $\alpha < 0$ [5, Corollary 4.3 (i)], which is the literal Open Question 2 for those spaces. Nor does it decide the behaviour at the point $\zeta = 1$ where f vanishes, nor regions tangent to $\mathbb{T}$ at ζ which contain no sector. The proof works for every choice $\varepsilon_m = M_m^{-\gamma}$ with $0 < \gamma < 1/2$ for the inward directions and $1/2 < \gamma < 1$ for the outward ones; the endpoint $\gamma = 1/2$ is where $|P_0| \asymp 1$ and Lemma 2 does not apply.

Problem 1. *Suppose that $\alpha < 0$. Are there a cyclic function $f \in D_\alpha$ and a point $\zeta \in \mathbb{T}$ such that no sector with vertex at ζ is devoid of zeros of the opas of $1/f$?*

The collision mechanism. The computation in the proof of Theorem 3 applies to clusters of any size. Suppose that $k \geq 2$ spurious zeros of φ_n cluster at a point $\xi \neq 0$ on the scale $1/n$, in the sense that $L(\xi + u/N)$, suitably normalized, tends to a monic polynomial $\mathcal{P}(u)$ of degree k, and that the remaining factors stay away from zero. Then the argument above shows that the partner equation (15) tends to $\mathcal{P}(u) - \xi\mathcal{P}'(u)$, while the conjectured pairing would require the zeros of $\mathcal{P}(u-\xi)$. If $\mathcal{P}(u) = u^k + c_1u^{k-1} + c_2u^{k-2} + \cdots$, the coefficients of u^{k-2} in these two polynomials are $c_2 - (k-1)c_1\xi$ and $c_2 - (k-1)c_1\xi + k(k-1)\xi^2/2$, so they never agree. For $\mathcal{P}(u) = u^k$ one gets $u^{k-1}(u - k\xi)$: $k-1$ partner zeros stay with the cluster and one is displaced by $k\xi/n$. The sum of the displacements is always $k\xi/n$, as predicted. Our proof realizes the case $k = 2$ for a fixed weight.

The counterexample concerns the displacement only. In the example the two sets have the same number of elements, and we do not know whether this is always so.

Problem 2. *Let f be a critical polynomial. Is the number of spurious zeros of φ_n equal, for all large n, to the number of zeros of S_n in $\mathbb{D}$ which do not accumulate on $\mathbb{T}$?*

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